

Constraints

Primary key constraint

Tuple cannot have duplicate PK.

Also, not NULL!

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
~~Widget~~	~~$15.99~~	~~GizmoWorks~~

Candidate key constraint

Tuple cannot have duplicate key.

However, allowed to be NULL.

fname	lname	netid	classification	phone	birthdate
John	Greiner	greiner	Faculty	713-348-3838	02/12/1967
Lynda	Bull	lb99	Student	713-555-6789	05/12/2001
Derek	Jackson	dj66	Student	913-555-1234	03/23/2000
Pam	Brown	pb21	Student	234-555-2398	07/21/2002
Sean	Harrington	sh84	Student	832-555-3461	02/13/2000
Jeff	Crooks	jc74	Student	832-555-7123	03/06/2001

Foreign key constraint

FK value must be one of the referent key values.

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL
Thingy	$105.99	ThingMaker

Manufacturer

man_name	man_address	man_city	man_state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA

Application – Lookup table

Lookup table

Ref. integrity ensures man_state is in this list.

State

state
AL
…
TX
WV

Manufacturer

man_name	man_address	man_city	man_state
GizmoWorks	123 Gizmo St.	Houston	TX
WidgetsRUs	20 Main St.	New York	NY
Hyper	1 Mission Dr.	San Francisco	CA
Gadgets	10 Minimax St.	Houston	TX
Wowza	99 Whynot Rd.	Palo Alto	CA

Not NULL constraint

Attribute must have a value.

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

Generic tuple constraint

Enforce any Boolean condition on each tuple.

E.g., prod_price > 0

Product

prod_name	prod_price	prod_manufacturer
Gizmo	$19.99	GizmoWorks
Powergizmo	$39.99	GizmoWorks
Widget	$19.99	WidgetsRUs
HyperWidget	$203.99	Hyper
VaporWare	NULL	NULL

Default value

Specify a default value for attribute.

Product

prod_name	prod_price	prod_manufacturer	prod_units_sold
Gizmo	$19.99	GizmoWorks	3000
Powergizmo	$39.99	GizmoWorks	250
Widget	$19.99	WidgetsRUs	2000
HyperWidget	$203.99	Hyper	1000
VaporWare	NULL	NULL	0